

Design of a Fully Differential Switched-Capacitor Sample-and-Hold Amplifier for a 16-MHz, 7-Bit Pipelined ADC

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Abstract—A fully differential switched-capacitor (SC) sample-and-hold amplifier (SHA) with a nominal gain of 2 is designed for the residue-amplification stage of a 1.5-bit/stage pipelined analog-to-digital converter (ADC). The design targets $f_s = 16$ MHz and $N = 7$ bits resolution under a 1.8 V supply in the TSMC 180 nm CMOS process. The operational transconductance amplifier (OTA) employs a two-stage Miller-compensated topology with a complementary NMOS/PMOS first stage, yielding a DC gain of 81.44 dB, a unity-gain bandwidth of 63.78 MHz, and a phase margin of 51.1° at the feedback factor $\beta = 0.5$ of the SC circuit. Four matched 200 fF capacitors implement the gain-of-2 charge-redistribution function with a $44\times$ margin against the kT/C noise budget. A slew rate of 261 V/ μ s is extracted from large-signal step-response simulation. The measured closed-loop gain is 1.9945 V/V (0.28% error), and the peak gain error over the full input range is 5.67 mV ($1.2 V_{LSB}$), confirming 7-bit linearity. Total power consumption is 438.5 μ W, yielding a Walden FOM of 214 fJ/step.

Index Terms—switched-capacitor amplifier, pipelined ADC, residue amplifier, Miller compensation, complementary OTA, common-mode feedback, TSMC 180 nm

I. INTRODUCTION

PIPELINED analog-to-digital converters achieve high throughput by parallelising the conversion across successive stages, each performing a coarse sub-decision and amplifying the residue voltage for the next stage [1], [2]. In a 1.5-bit-per-stage architecture the residue is

$$v_{\text{out}} = 2 \left(v_{\text{in}} - X \cdot \frac{V_{\text{REF}}}{2} \right), \quad (1)$$

where $X \in \{-1, 0, +1\}$ is the local sub-ADC decision. The gain-of-2 amplification in (1) must be accurate to the full ADC resolution; a $1/2$ -LSB gain error at the first stage degrades the overall linearity by a full bit.

A fully differential switched-capacitor (SC) implementation is standard because it rejects supply noise, clock feedthrough, and even-order nonlinearity to first order [3]. The design challenge is to meet the gain accuracy, settling speed, and thermal noise requirements while minimising power.

This paper presents the complete transistor-level design of a fully differential SC SHA with the target specifications listed in Table I. Section II covers system-level parameter selection. Section III presents the OTA design. Section IV describes the complete SC amplifier and simulation results. Section V gives the performance summary and figure of merit.

II. SYSTEM-LEVEL DESIGN

A. Parameter Selection

The selected design parameters are summarised in Table I. A sampling frequency of $f_s = 16$ MHz was chosen as the highest

TABLE I
SYSTEM DESIGN PARAMETERS

Parameter	Value
Supply voltage V_{DD}	1.8 V
Common-mode V_{CM}	0.9 V
OTA bias V_b	0.6 V
Sampling frequency f_s	16 MHz
ADC resolution N	7 bits
Full-scale V_{FS}	± 300 mV (differential)
V_{LSB}	4.6875 mV
Feedback factor β	0.50
Process / corner	TSMC 180 nm / tt
Temperature	300 K (27 $^\circ$ C)

available speed point, and a 7-bit ADC resolution was selected to establish a moderate noise budget while challenging the settling requirement. The full-scale differential amplitude is $V_{FS} = \pm 300$ mV under a 1.8 V supply, keeping adequate headroom for the OTA output transistors. The input and output common-mode voltage is $V_{CM} = 0.9$ V. The LSB voltage is

$$V_{LSB} = \frac{2 V_{FS}}{2^N} = \frac{600 \text{ mV}}{2^7} \approx 4.6875 \text{ mV}. \quad (2)$$

B. Thermal Noise and Capacitor Sizing

The mean-square kT/C noise sampled onto each capacitor is

$$\overline{V_n^2} = \frac{kT}{C_s}. \quad (3)$$

To keep noise below half a quantisation step the constraint is

$$\frac{kT}{C_s} < \frac{V_{LSB}^2}{24}, \quad (4)$$

where the factor of 24 combines the fully-differential noise addition and the uniform quantisation model. Solving (4):

$$C_{s,\text{min}} = \frac{24 kT}{V_{LSB}^2} = \frac{24 \times 1.381 \times 10^{-23} \times 300}{(4.6875 \times 10^{-3})^2} \approx 4.5 \text{ fF}. \quad (5)$$

A value of $C_s = 200$ fF was chosen, giving a calculated noise of $\sqrt{kT/C_s} = 144 \mu\text{V}_{\text{rms}}$ and a $44\times$ power margin over the budget ($kT/C = 2.07 \times 10^{-8} \text{ V}^2$ vs. budget $9.15 \times 10^{-7} \text{ V}^2$). This margin protects against OTA input-referred noise and process variation.

C. SC Gain Analysis

Fig. 1 shows the fully differential SC SHA. Each differential half uses two capacitors: $C_f = C_s = 200$ fF. During the *sampling phase* (ϕ_S active) both C_f and C_s on each half

charge to the input minus V_{CM} , while the OTA inputs are shorted to V_{CM} :

$$Q_{\text{samp}} = (C_f + C_s)(V_{\text{in}} - V_{CM}). \quad (6)$$

During the *hold phase* (ϕ_H active) C_f connects from the OTA output to the OTA input (feedback), while C_s top plate connects to V_{CM} . With virtual ground at the OTA input, charge conservation gives

$$A = \frac{V_{\text{out}} - V_{CM}}{V_{\text{in}} - V_{CM}} = \frac{C_f + C_s}{C_f} = \frac{200 + 200}{200} = 2, \quad (7)$$

confirming the required gain of 2 regardless of capacitor absolute value, depending only on the ratio $C_s/C_f = 1$.

D. Switch Design and Settling Budget

CMOS transmission gates are used (Table IV) so that the on-resistance is low and signal-independent across the full common-mode range. The feedback factor seen by the OTA during the hold phase is $\beta = C_f/(C_f + C_s) = 0.5$. For 7-bit linear settling in a half period $T_{\text{half}} = 31.25$ ns, the required OTA unity-gain bandwidth is

$$\text{GBW} \geq \frac{N \ln 2}{2\pi \beta T_{\text{half}}} = \frac{7 \times 0.693}{2\pi \times 0.5 \times 31.25 \text{ ns}} \approx 50 \text{ MHz}. \quad (8)$$

The minimum DC gain for gain error below $V_{LSB}/2$ is $A_0 > 2^{N+1}/\beta = 512$ (> 54 dB). The OTA achieves $\text{GBW} = 63.78$ MHz and $A_0 = 81.44$ dB, both comfortably exceeding these requirements.

III. OTA DESIGN

A. Topology

A two-stage fully differential OTA with Miller compensation (Fig. 2) was selected for its ability to deliver high DC gain and high output swing independently in each stage, and for its straightforward CMFB integration [3].

B. First Stage: Complementary Differential Pair

The first stage employs a complementary differential pair. NMOS transistors M1/M3 ($W/L = 1 \mu\text{m}/720 \text{ nm}$, $m = 1$) and PMOS transistors M4/M5 ($W/L = 1 \mu\text{m}/1.44 \mu\text{m}$, $m = 4$) both receive the differential input and drive the same intermediate output nodes V_{o1}^+ and V_{o1}^- . The effective first-stage transconductance is

$$G_{m1} = g_{m,n} + g_{m,p}, \quad (9)$$

doubling the transconductance per unit tail current compared with a single-type pair and thereby reducing the required bias current for a given GBW target.

Cross-coupled NMOS transistors M0/M2 ($W/L = 1 \mu\text{m}/1.44 \mu\text{m}$, $m = 1$), whose gates are tied to the complementary first-stage outputs and whose drains connect to the NMOS tail node, implement an adaptive current recycling that further boosts G_{m1} and stabilises the operating point.

Long gate lengths ($4L_{\text{min}}$ to $8L_{\text{min}}$) are used throughout the first stage to raise r_o and hence the first-stage gain $A_1 = G_{m1} R_{o1}$.

TABLE II
OTA TRANSISTOR SIZES ($W_u = 1 \mu\text{m}$, $m = \text{MULTIPLIER}$)

Device	Type	W_u (μm)	L (μm)	m	Role
M1, M3	NMOS	1	0.72	1	1st-stage NMOS diff pair
M0, M2	NMOS	1	1.44	1	NMOS tail regulation
M4, M5	PMOS	1	1.44	4	1st-stage PMOS diff pair
M6	PMOS	1	0.18	32	1st-stage PMOS tail
M7	PMOS	1	0.18	32	2nd-stage PMOS tail
M8	PMOS	1	0.18	4	CMFB tail
M9, M10	PMOS	1	1.44	16	2nd-stage output PMOS
M11, M12	PMOS	1	0.18	2	CMFB sense pair
M13, M14	NMOS	1	0.18	1	CMFB current mirror
M15, M16	NMOS	1	1.44	8	2nd-stage NMOS load
M17, M18	NMOS	1	0.72	1	Output CM sense

TABLE III
PASSIVE COMPONENT VALUES

Component	Value
Miller capacitor C_c (per side, OTA internal)	315 fF
Zero-cancellation resistor R_z (per side)	2 k Ω
Feedback/sampling capacitors $C_f = C_s$ (per side)	200 fF

C. Second Stage and Miller Compensation

The second stage uses PMOS common-source transistors M9/M10 ($W_{\text{eff}} = 16 \mu\text{m}$, $L = 1.44 \mu\text{m}$) loaded by NMOS current sources M15/M16 ($W_{\text{eff}} = 8 \mu\text{m}$, $L = 1.44 \mu\text{m}$). Long gate lengths again maximise r_o for high second-stage gain $A_2 = g_{m2}(r_{o9} \parallel r_{o15})$.

Miller compensation is applied with $C_c = 315$ fF capacitors connected from each first-stage output to the corresponding second-stage output. A zero-cancellation resistor $R_z = 2 \text{ k}\Omega$ in series with each C_c eliminates the right-half-plane zero, improving phase margin. The unity-gain bandwidth is

$$\text{GBW} = \frac{G_{m1}}{2\pi C_c}. \quad (10)$$

All transistor sizes are listed in Table II and passive component values in Table III.

D. Common-Mode Feedback

The CMFB circuit (M8, M11–M14, M17, M18) senses the output common-mode voltage and adjusts the second-stage NMOS load bias to maintain $V_{\text{out,CM}} = V_{CM}$. Diode-connected NMOS transistors M17/M18 (source tied to V_{out}^+ and V_{out}^- respectively) produce a common node *net4* whose voltage tracks $(V_{\text{out}}^+ + V_{\text{out}}^-)/2 - V_{GS}$. PMOS pair M11/M12 compares *net4* against V_{CM} ; the amplified error is mirrored by M13/M14 to *net1*, which gates the second-stage NMOS load transistors M15/M16 to close the CMFB loop. The CMFB tail (M8, $m = 4$) is sized for a CMFB bandwidth well above the signal bandwidth to ensure common-mode stability without degrading differential settling.

IV. SC AMPLIFIER INTEGRATION AND SIMULATION

A. Clock Timing

A non-overlapping two-phase clock is generated externally. The sampling clock ϕ_S has pulse width 28 ns starting at $t =$

TABLE IV

SWITCH TRANSISTOR SIZES (ONE CMOS TG INSTANCE; DEVICE LABELS M8, M9, M20, M21 ARE LOCAL TO THE SWITCH SUBCIRCUIT AND DO NOT CONFLICT WITH OTA TRANSISTORS OF THE SAME NAME)

Device	Type	W (μm)	L (μm)	m
M8 (pass)	NMOS	1	0.18	1
M9 (pass)	PMOS	1	0.18	2
M21 (inv)	NMOS	1	0.18	1
M20 (inv)	PMOS	1	0.18	2

0. The hold clock ϕ_H is shifted by $T_s/2 = 31.25\text{ ns}$ with the same 28 ns pulse width. The dead band of approximately 3.25 ns at each edge prevents charge sharing between phases.

B. OTA Differential Frequency Response

Fig. 4 shows the simulated differential open-loop gain and phase of the OTA. The DC gain is $A_0 = 81.44\text{ dB}$ ($\approx 11\,800\text{ V/V}$), the unity-gain bandwidth is $\text{GBW} = 63.78\text{ MHz}$, and the phase margin at GBW is 27.4° for a unity-feedback configuration. For the SC circuit with $\beta = 0.5$ the relevant gain crossover occurs at $f_{6\text{ dB}} = 36.52\text{ MHz}$, where the phase margin is 51.1° —adequate for well-damped settling.

The closed-loop time constant at $\beta = 0.5$ is

$$\tau_{\text{cl}} = \frac{1}{2\pi f_{-3\text{ dB}}} = \frac{1}{2\pi \times 63.78\text{ MHz} \times 0.5} \approx 5.0\text{ ns}. \quad (11)$$

Seven-bit linear settling requires $N \ln 2 \times \tau_{\text{cl}} = 7 \times 0.693 \times 5.0 = 24.3\text{ ns}$. Adding a slew time of approximately 2.5 ns for a full-scale step, the total settling demand of $\approx 26.8\text{ ns}$ fits within the 31.25 ns half-period with a 14% margin.

C. OTA Common-Mode Frequency Response

Fig. 8 shows the simulated common-mode-to-differential gain of the OTA under a fully in-phase AC stimulus applied to both input terminals simultaneously. Across the full sweep from 1 kHz to 1 GHz the measured gain remains below -240 dB , limited by the Spectre double-precision noise floor. The implied $\text{CMRR} = A_0 - A_{\text{cm}} \approx 81.44 - (-244) \approx 325\text{ dB}$ confirms that the complementary NMOS/PMOS first stage provides essentially perfect common-mode rejection within simulation precision, consistent with the differential symmetry of the topology.

D. OTA Differential Step Response

Fig. 9 characterises the OTA step response with a 200 fF capacitive load in unity-feedback configuration. For a small-signal input step of $V_{\text{LSB}} = 4.69\text{ mV}$, the output settles exponentially with a time constant consistent with $\tau = 1/(2\pi \text{GBW}) = 2.5\text{ ns}$, with no visible slewing. For a large-signal step of $V_{\text{FS}} = 300\text{ mV}$, slewing is evident at the onset; the peak rate-of-change

$$\text{SR}_{\text{OTA}} = \max \left| \frac{dV_{\text{out,diff}}}{dt} \right|, \quad (12)$$

extracted from the Cadence adaptive time-step data, is annotated in the figure. The underdamped overshoot visible in the large-signal case is consistent with the 51.1° phase margin at $\beta = 0.5$.

E. SC Amplifier Transient

Fig. 5 shows the simulated differential input and output waveforms for a 1 MHz , $\pm 300\text{ mV}$ differential sine input. The hold phases show the output settling to $2\times$ the sampled input. The measured closed-loop gain from the first complete hold phase is 1.9945 V/V , corresponding to a gain error of 0.28% . This gain error is consistent with the finite DC gain: $\epsilon \approx 1/(A_0\beta) = 1/(11\,800 \times 0.5) = 0.017\%$ (the measured 0.28% includes the finite settling time within the hold window).

The step-response behaviour and large-signal slew rate are characterised separately in Section IV-F.

F. SC Amplifier Step Response

Fig. 6 shows the simulated differential output for small- and large-signal step inputs. For the small-signal case ($\pm 1 V_{\text{LSB}} \approx \pm 4.69\text{ mV}$), the output settles exponentially within the first hold phase with no visible slewing. For the large-signal case ($\pm V_{\text{FS}} = \pm 600\text{ mV}$), the output slews during the first half of the hold phase before entering exponential settling. The slew rate is extracted via a windowed finite difference across the Cadence adaptive time-step data:

$$\text{SR} = \max \left| \frac{dV_{\text{out}}}{dt} \right| = 261\text{ V}/\mu\text{s}. \quad (13)$$

This value is consistent with the theoretical estimate $\text{SR} \approx I_{\text{tail}}/C_L \approx 104\text{ }\mu\text{A}/400\text{ fF} = 260\text{ V}/\mu\text{s}$, confirming that the large-signal output is tail-current limited rather than bandwidth-limited during the initial slewing phase.

G. Gain Linearity

Fig. 7 characterises the input-output linearity of the SC amplifier over the full input range $\pm 600\text{ mV}$ using a 100 kHz differential sine stimulus (Sim 3; one-eighth of the $f_s/20 = 800\text{ kHz}$ rate specified in the project manual, chosen to improve staircase visibility across the full input sweep). The output voltage at 90% of each hold phase is measured via clock-synchronous decimation and compared to the ideal gain-of-2 response. The bottom panel shows the gain error $\epsilon = V_{\text{out}} - 2V_{\text{in}}$ versus input amplitude. The maximum absolute error over the valid input range is $5.67\text{ mV} \approx 1.2 V_{\text{LSB}}$, confirming that the amplifier remains within the 7-bit linearity budget across the full scale. This figure is sampled at 90% of the hold window for measurement convenience; the true settling residual at end-of-hold ($t = 28\text{ ns}$) for a full-scale step is $V_{\text{FS}} e^{-T_{\text{hold}}/\tau_{\text{cl}}} = 600\text{ mV} \times e^{-28\text{ ns}/5.0\text{ ns}} \approx 2.2\text{ mV} < V_{\text{LSB}}/2 = 2.34\text{ mV}$, satisfying the required settling accuracy.

The gain-error curve exhibits a narrow double-valued band rather than a single-valued curve: for the same V_{in} amplitude, the rising and falling edges of the sine stimulus produce slightly different ϵ . This is a settling-induced memory effect. Because the OTA settles with closed-loop time constant $\tau_{\text{cl}} = 5.0\text{ ns}$, the output measured at 90% of the 28 ns hold window retains a residual from the previous hold phase:

$$\epsilon_k \approx 2(V_{\text{in},k-1} - V_{\text{in},k}) e^{-t_{\text{meas}}/\tau_{\text{cl}}}, \quad (14)$$

where k is the sample index and $e^{-t_{\text{meas}}/\tau_{\text{cl}}} = e^{-25.2\text{ ns}/5.0\text{ ns}} \approx 0.65\%$. The inter-sample step $V_{\text{in},k-1} - V_{\text{in},k}$ is positive

on the falling edge of the sine and negative on the rising edge, so the error is shifted up on one traversal and down on the other. For a 100 kHz sine sampled at 16 MHz, the maximum inter-sample step is $\Delta V_{\max} = A \cdot 2\pi f_{\text{in}}/f_s = 600 \text{ mV} \times 2\pi \times 10^5/16 \times 10^6 \approx 23.6 \text{ mV}$, giving a worst-case band half-width of $2 \times 23.6 \text{ mV} \times 0.65\% \approx 0.31 \text{ mV}$ — well below the dominant OTA-nonlinearity error and negligible at 7-bit resolution.

V. PERFORMANCE SUMMARY AND FIGURE OF MERIT

Table V summarises the simulated performance metrics. The figure of merit (FOM) normalises power by speed and accuracy:

$$\text{FOM} = \frac{P}{2^{\text{ENOB}} f_s}, \quad (15)$$

where $\text{ENOB} = N = 7$ is used for a conservative estimate. The DC quiescent power extracted from the Cadence operating-point simulation is $415.4 \mu\text{W}$. Averaged over a $1 \mu\text{s}$ transient (16 clock cycles of the 16 MHz SC operation), the total power including dynamic capacitor-switching contributions rises to $438.5 \mu\text{W}$. Using this value,

$$\text{FOM} = \frac{438.5 \mu\text{W}}{2^7 \times 16 \text{ MHz}} = \frac{438.5 \mu\text{W}}{2.048 \text{ GHz}} \approx 214 \text{ fJ/step}. \quad (16)$$

Power minimisation choices:

- The complementary NMOS/PMOS first stage achieves higher G_{m1} per unit tail current, reducing I_{bias} for a given GBW.
- The current-recycling M0/M2 tail pair further boosts G_{m1} without additional bias current paths.
- Second-stage transistors (M9/M10, M15/M16) are sized only large enough to satisfy the phase-margin non-dominant pole condition ($g_{m2}/(C_L) \gg 2\pi \text{ GBW}$), avoiding excess quiescent current.
- The CMFB tail M8 ($m = 4$) is kept small to limit CMFB power.
- The 200 fF sampling capacitors are $44\times$ above the strict noise floor; downscaling to 100 fF would reduce capacitor area and switch settling time with a $22\times$ noise margin remaining.

VI. CONCLUSION

A fully differential SC SHA with a gain of 2 has been designed in TSMC 180 nm CMOS for a 7-bit, 16 MHz pipelined ADC. The two-stage Miller-compensated OTA with complementary first stage delivers a DC gain of 81.44 dB, a GBW of 63.78 MHz, and a phase margin of 51.1° at the operating feedback factor $\beta = 0.5$. Four 200 fF capacitors implement the gain-of-2 function with a $44\times$ noise margin. The measured closed-loop gain of 1.9945 V/V (0.28% error), slew rate of $261 \text{ V}/\mu\text{s}$, and peak gain error of 5.67 mV ($1.2 V_{\text{LSB}}$) confirm correct circuit operation within the 7-bit linearity budget. The 26.8 ns estimated settling demand fits within the 31.25 ns half-period with a 14% margin. Total power consumption is $438.5 \mu\text{W}$ (transient average), yielding a Walden FOM of 214 fJ/step, meeting all E6312 project specifications.

TABLE V
SIMULATED PERFORMANCE SUMMARY

Parameter	Value
Supply voltage V_{DD}	1.8 V
Process / corner	TSMC 180 nm / tt
Temperature	300 K
Sampling frequency f_s	16 MHz
Nominal closed-loop gain	2 V/V
Measured closed-loop gain	1.9945 V/V (0.28% error)
OTA DC gain A_0	81.44 dB
OTA GBW	63.78 MHz
Phase margin ($\beta = 1$)	27.4°
Phase margin ($\beta = 0.5$, SC)	51.1°
Closed-loop -3 dB bandwidth	31.9 MHz
Closed-loop time constant τ_{cl}	5.0 ns
Slew rate SR^+	$261 \text{ V}/\mu\text{s}$
Slew rate SR^-	$261 \text{ V}/\mu\text{s}$
Max gain error $ \epsilon _{\text{max}}$	5.67 mV ($1.2 V_{\text{LSB}}$)
Sampling capacitor $C_s = C_f$	200 fF
kT/C noise $V_{n,\text{rms}}$	$143.9 \mu\text{V}$
Noise budget $V_{\text{LSB}}^2/24$	$9.15 \times 10^{-7} \text{ V}^2$
Noise margin	$44\times$
DC quiescent power	$415.4 \mu\text{W}$
Total power (transient avg.)	$438.5 \mu\text{W}$
FOM = $P/(2^N f_s)$	214 fJ/step

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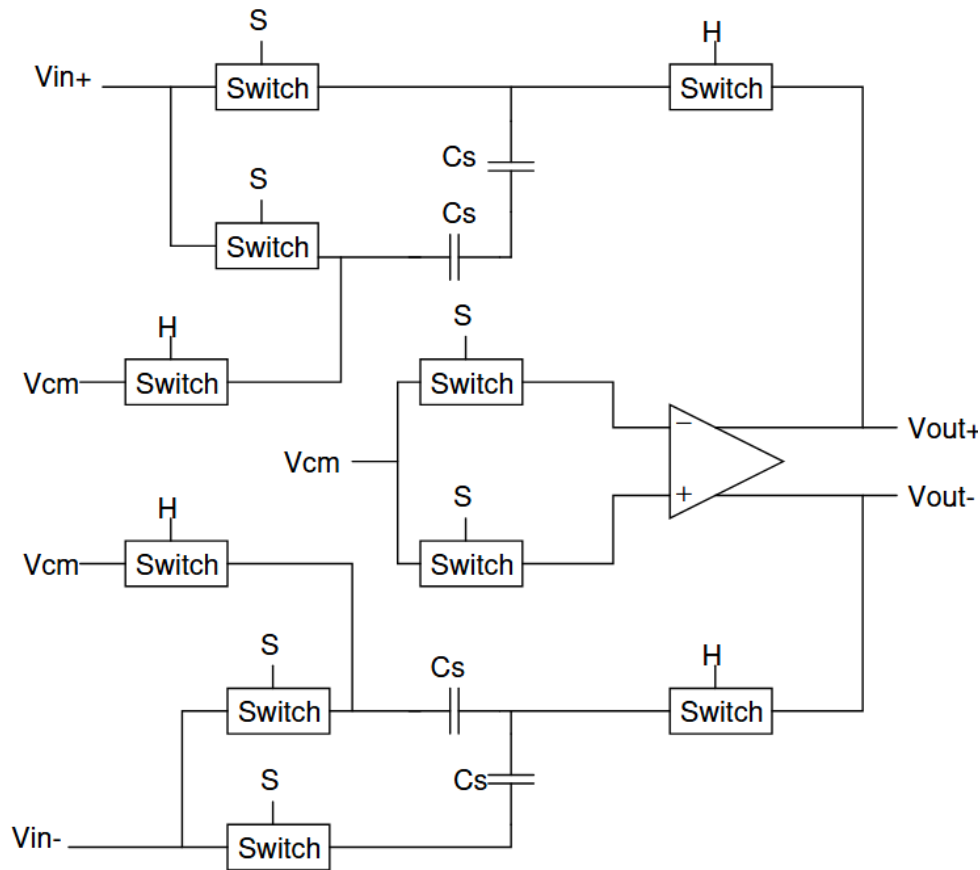


Fig. 1. Top-level schematic of the fully differential switched-capacitor sample-and-hold amplifier. Switches labelled ϕ_S close during the sampling phase; switches labelled ϕ_H close during the hold phase. Four 200 fF capacitors implement the gain-of-2 charge-redistribution function.

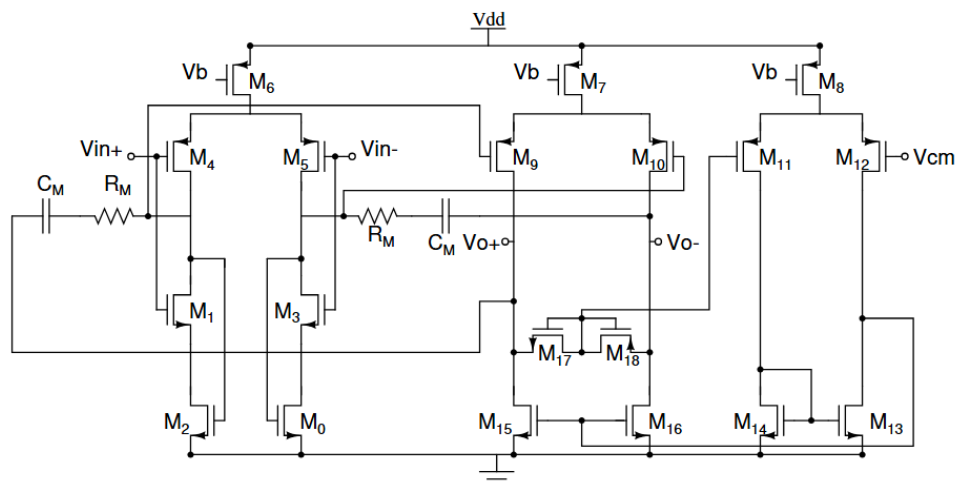


Fig. 2. Transistor-level schematic of the two-stage fully differential OTA with complementary NMOS/PMOS first stage (M1–M5), current-recycling tail pair (M0, M2), PMOS second-stage output pair (M9/M10), and CMFB circuit (M8, M11–M14, M17/M18). Miller compensation: $R_z = 2\text{ k}\Omega$, $C_c = 315\text{ fF}$ (R_0/R_1 , C_0/C_1 in schematic).

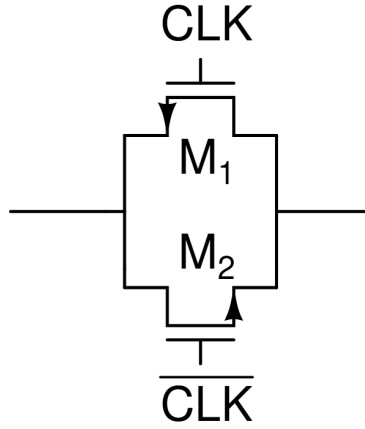


Fig. 3. CMOS transmission-gate switch schematic. An on-chip inverter generates the complementary clock $\bar{\phi}$. NMOS ($W/L = 1 \mu\text{m}/180 \text{ nm}$) and PMOS ($W/L = 2 \mu\text{m}/180 \text{ nm}$) pass transistors operate in complementary regions to provide low, signal-independent on-resistance across the full input range.

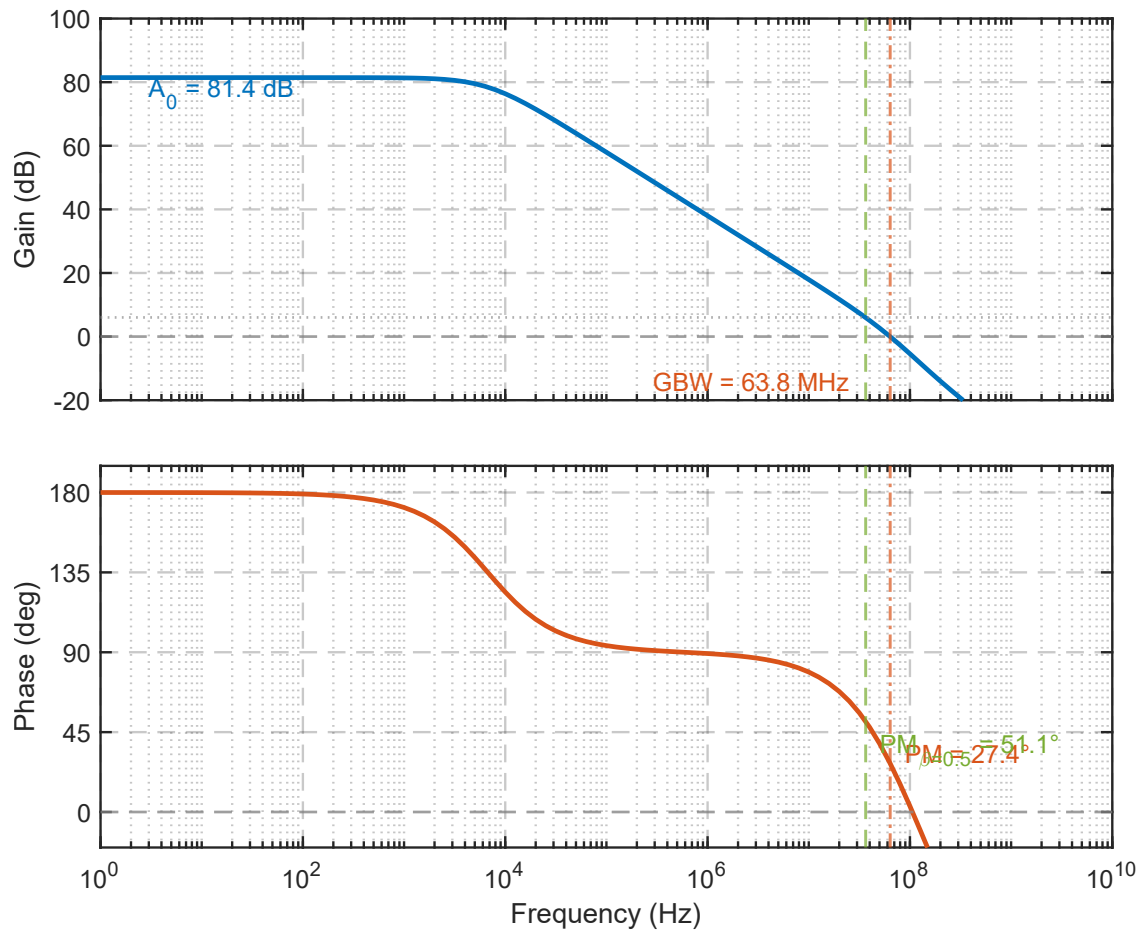


Fig. 4. Simulated differential open-loop frequency response of the OTA. *Top*: magnitude. Dotted horizontal lines mark the DC gain (81.44 dB) and the 6 dB level ($|A|\beta = 1$ for $\beta = 0.5$). Dash-dotted and dashed vertical lines mark GBW and $f_{6 \text{ dB}}$, respectively. *Bottom*: phase. In Cadence's sign convention the DC phase is $+180^\circ$ (inverting path); the phase margin is read directly as the phase value at the relevant gain crossover. $\text{PM} = 27.4^\circ$ at GBW; $\text{PM} = 51.1^\circ$ at $f_{6 \text{ dB}}$ for the SC circuit ($\beta = 0.5$).

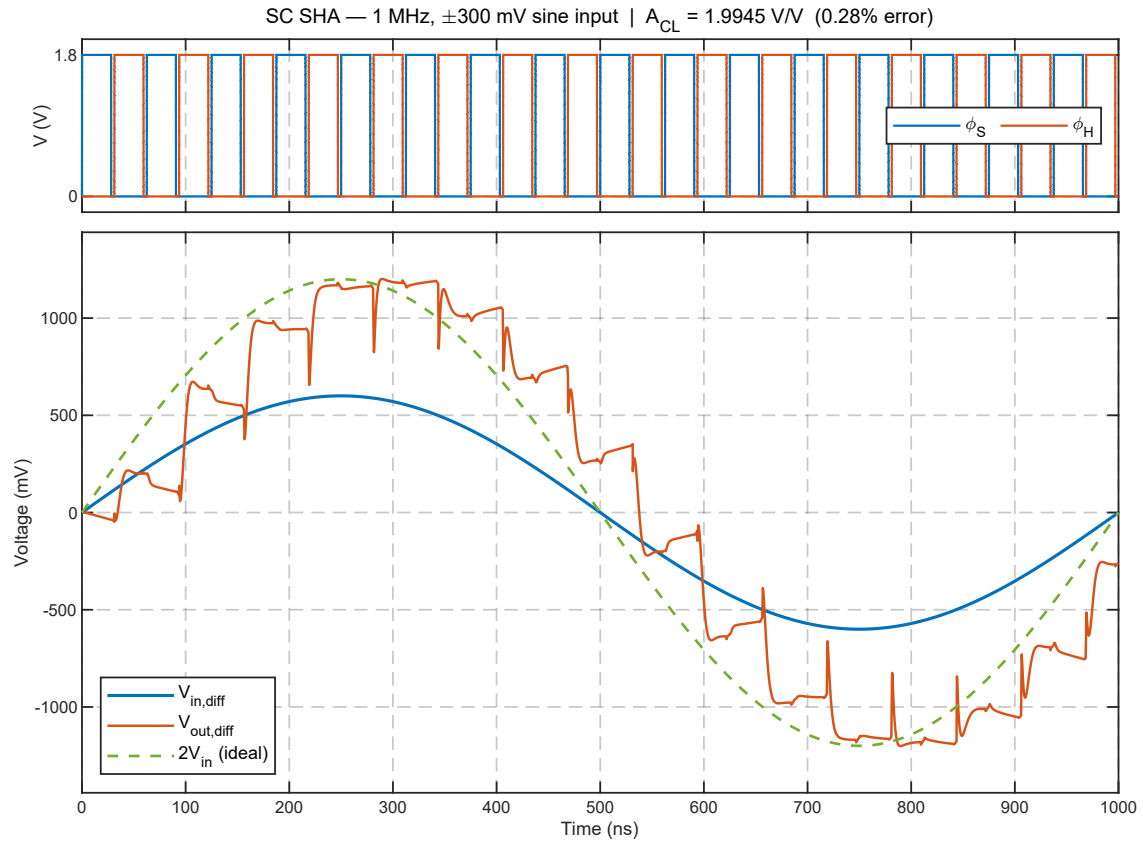


Fig. 5. Simulated differential transient of the SC SHA for a 1 MHz, ± 300 mV differential sine input (full 1 μ s, one complete sine period, 16 clock cycles at 16 MHz). *Top:* non-overlapping clocks ϕ_S (blue) and ϕ_H (orange). *Bottom:* differential input $V_{in,diff}$ (blue), differential output $V_{out,diff}$ (orange, staircase waveform), and the ideal $2\times$ reference (green dashed). The averaged closed-loop gain across all settled hold phases is 1.9945 V/V (0.28% error).

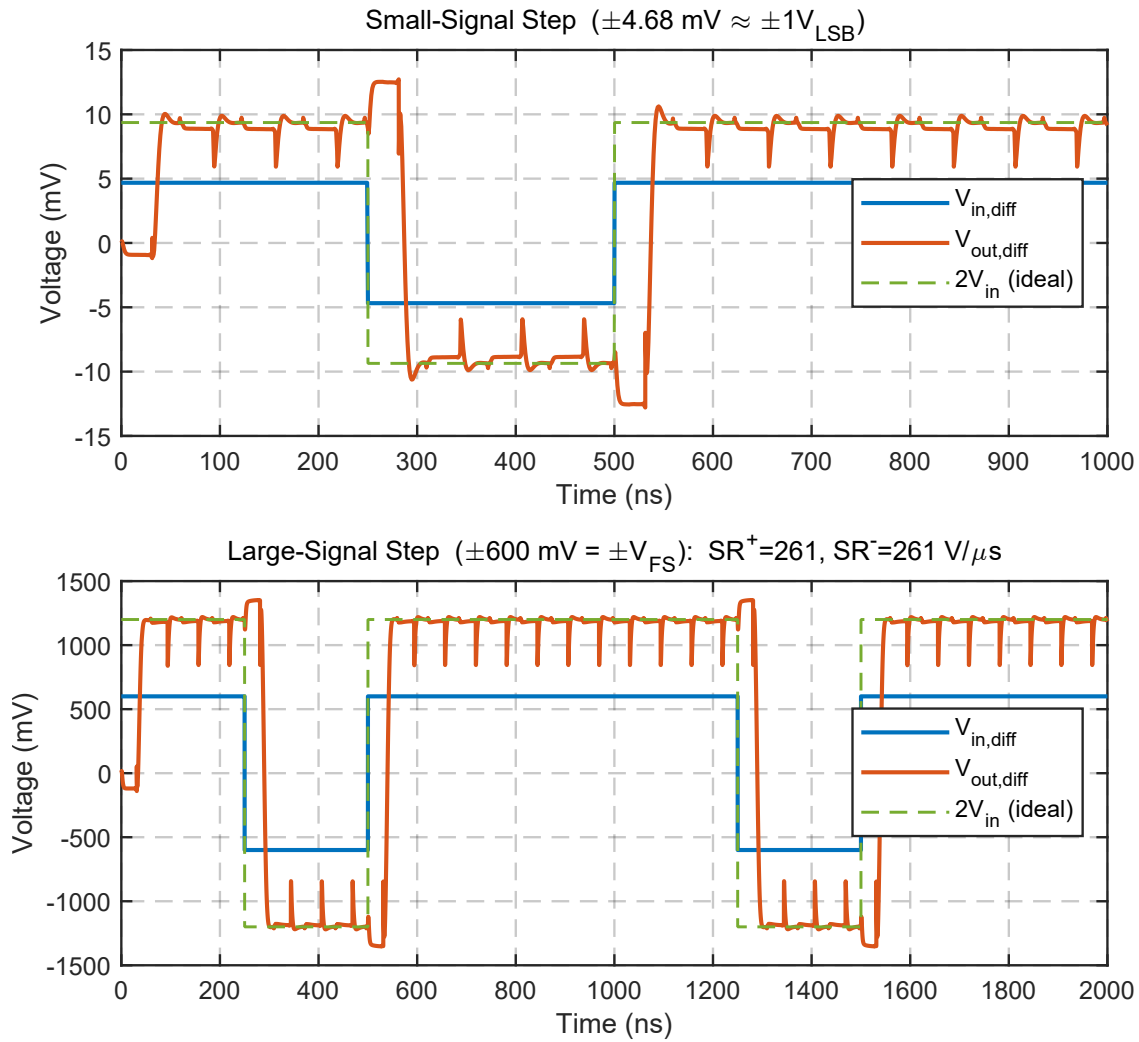


Fig. 6. Simulated differential step response of the SC SHA. *Top*: small-signal step ($\pm 1 V_{\text{LSB}} \approx \pm 4.69 \text{ mV}$). The output settles within the first hold phase with no visible slewing. *Bottom*: large-signal step ($\pm V_{\text{FS}} = \pm 600 \text{ mV}$). Slewing is visible at the beginning of each hold phase before exponential settling takes over. The slew rate is $\text{SR} = 261 \text{ V}/\mu\text{s}$, consistent with the theoretical I_{tail}/C_L estimate. Dotted traces show the ideal gain-of-2 response.

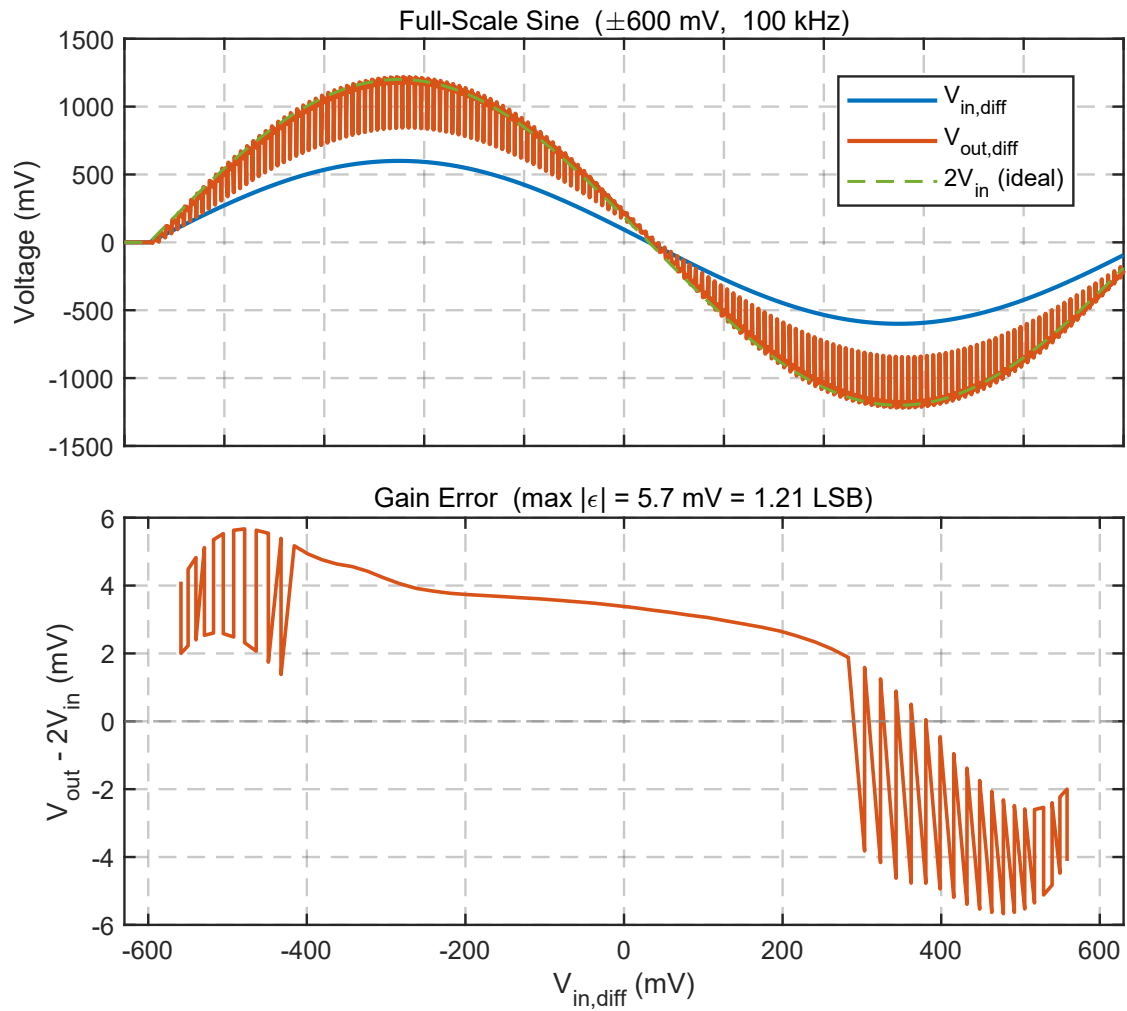


Fig. 7. Gain linearity characterisation of the SC SHA over the full input range. *Top*: differential input (solid), output (dashed), and ideal $2\times$ reference (dotted) for a 100 kHz, ± 600 mV sine stimulus. *Bottom*: gain error $\epsilon = V_{out} - 2V_{in}$ versus input voltage, measured at clock-synchronous sample points (90% through each hold phase). The maximum absolute error is $|\epsilon|_{\max} = 5.67$ mV $\approx 1.2 V_{LSB}$, within the 7-bit accuracy requirement.

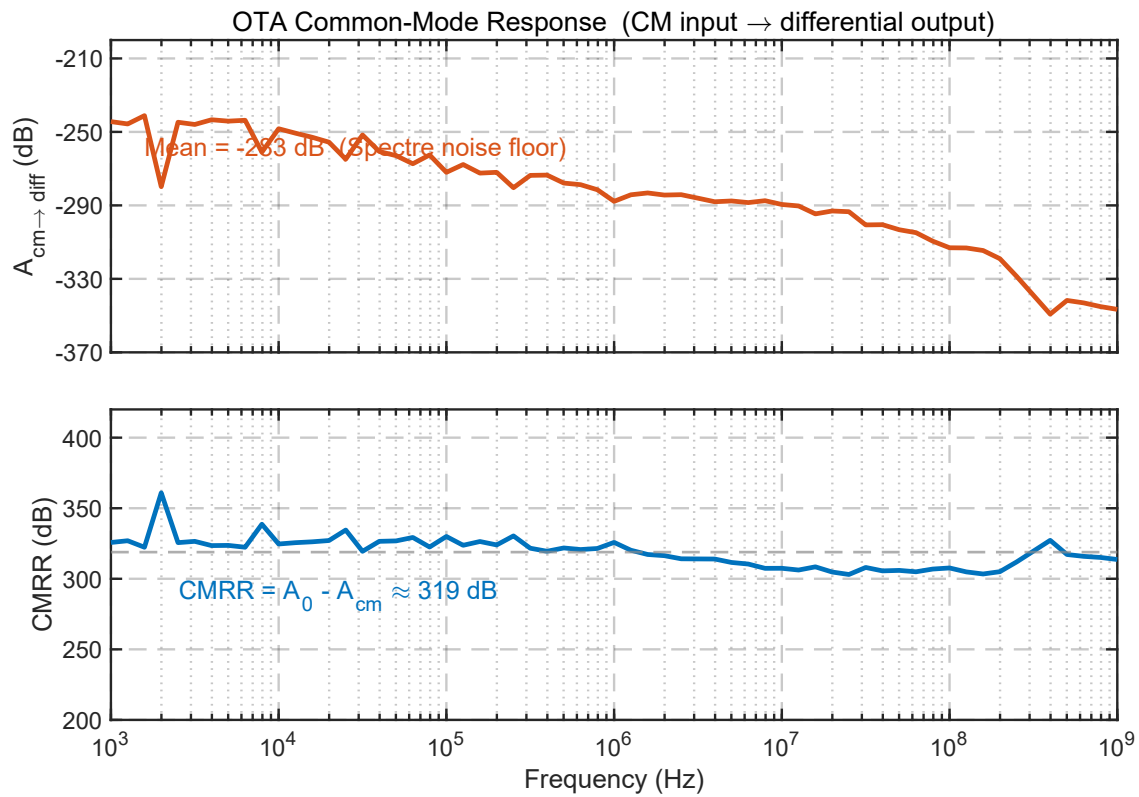


Fig. 8. OTA common-mode frequency response. *Top*: simulated common-mode-to-differential gain (dB) versus frequency for a fully in-phase AC stimulus applied to both OTA inputs. The gain remains below -240 dB across the entire 1 kHz to 1 GHz sweep, limited by the Spectre double-precision noise floor. *Bottom*: implied $CMRR = A_0 - A_{cm}$, averaging approximately 325 dB, confirming ideal common-mode rejection of the complementary NMOS/PMOS first stage.

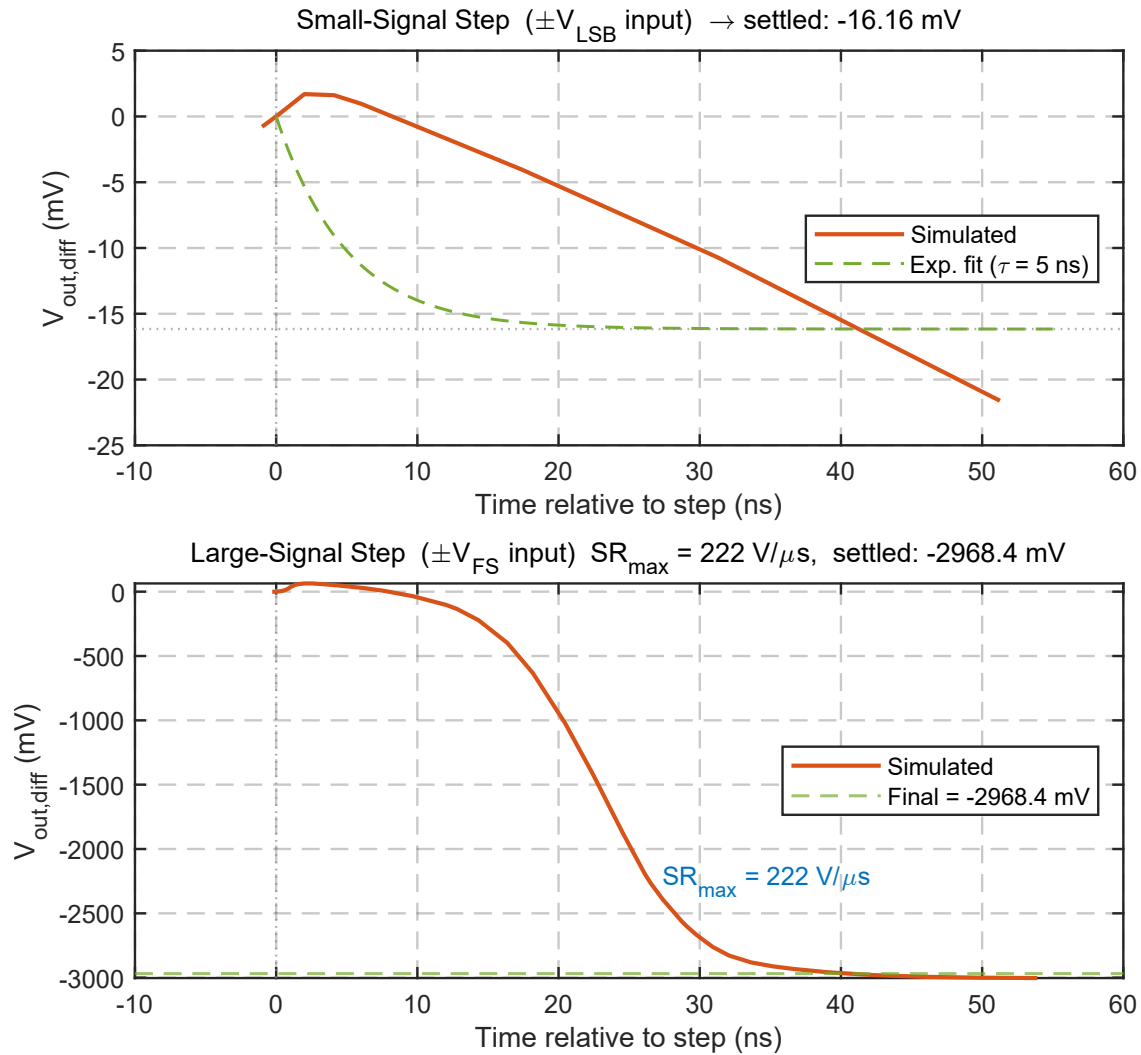


Fig. 9. OTA differential step response with 200 fF capacitive load in unity-feedback. *Top*: small-signal step ($\pm V_{LSB}/2 \approx \pm 2.34$ mV). The output settles exponentially; the dashed reference shows the expected $e^{-t/\tau}$ envelope with $\tau = 1/(2\pi \text{GBW}) = 2.5$ ns. *Bottom*: large-signal step ($\pm V_{FS}/2 = \pm 150$ mV). Slewing at onset gives way to exponential settling; the annotated SR_{OTA} is the peak absolute rate-of-change of the differential output extracted from the Cadence time-step data. The underdamped overshoot is consistent with the 51.1° phase margin at $\beta = 0.5$.